Data Flow Diagram (DFD Level 0 & 1)

The structural movement of data through the system follows a direct verification, preprocessing, and reporting timeline:

1. **Information Ingestion:** The network analyst or an external pipeline application submits a data array containing exactly 10 core attributes (such as duration, src_bytes, dst_bytes, and num_failed_logins).
2. **Request Resolution:** The **Flask Application Route Engine** intercepts the message bundle, validates structure formatting, and triggers the inference pipeline.
3. **Statistical Normalization:** The input feature collection passes into the **StandardScaler pipeline**, which transforms raw integers using a zero-mean and unit-variance configuration.
4. **Ensemble Execution:** The normalized data array is evaluated across the 100 decision trees of the pre-trained **Random Forest Classifier** to compute classification probabilities.
5. **Log & Return Broadcast:** The backend pushes the classified state label and its confidence percentage to the system's runtime log dashboard while instantly transmitting the prediction response back to the client.

2. **User Stories & Technical Roadmap**

1. Dashboard View (USN-1)

- **Who wants it:** The System Administrator (the person managing the network).
- **What it does:** Creates a central homepage dashboard (/dashboard) that shows colorful, live visual charts (using a tool called Chart.js).
- **Why it matters:** It lets the admin see the health of the entire network and notice any sudden spikes in cyberattacks instantly without digging through raw data.
- **When it's built:** Priority is **High**; built right away in **Sprint 1**.

2. Manual Inspection Form (USN-2)

- **Who wants it:** The Security Analyst (the cybersecurity expert inspecting threats).
- **What it does:** Provides a webpage form (/predict) with boxes where the analyst can manually type in 10 specific mathematical clues about a suspicious connection.
- **Why it matters:** When clicked, the AI reads these 10 clues and immediately outputs what kind of attack it is (like a DoS or a Probe attack) and how confident the AI is in its guess.

- **When it's built:** Priority is **High**; built right away in **Sprint 1**.

3. One-Click Quick Test Buttons (USN-3)

- **Who wants it:** The Security Analyst.
- **What it does:** Adds pre-configured "Demo Simulation" buttons directly onto the form page.
- **Why it matters:** Instead of manually typing out long numbers to test the system, the analyst can click a single button (like a "Simulate DoS Attack" button) to automatically fill out the form with a realistic attack profile.
- **When it's built:** Priority is **Medium**; built a little later in **Sprint 2**.

4. Incident Log History (USN-4)

- **Who wants it:** The System Administrator.
- **What it does:** Creates an operational log page (/alerts) that acts as a continuous history book.
- **Why it matters:** It records every single threat detected by the AI over time. It lists exactly when the threat happened (timestamps), what type of attack it was, and includes a visual progress bar showing the AI's confidence score.
- **When it's built:** Priority is **High**; built right away in **Sprint 1**.

5. Automation API Endpoint (USN-5)

- **Who wants it:** External Software (other applications or automated company scripts).
- **What it does:** Exposes a secure link backend path (/api/predict) specifically designed for computers to talk to other computers.
- **Why it matters:** It lets outside security apps silently send network data bundles into your system in the background and instantly receive a clean, structured reply stating whether the data contains a live threat.
- **When it's built:** Priority is **Medium**; built a little later in **Sprint 2**.